1 Singular Homology

1.1 The Construction

Definition 1.1.1

Let X be any topological space and $n \geq 0$ be an integer. By a ***singular n-simplex*** in X, we mean a continuous map $\sigma : |\Delta_n| \to X$. By a ***singular n-chain*** in X, we mean a finite sum $\sum_i n_i \sigma_i$, where $n_i \in \mathbb{Z}$ and σ_i are singular n-simplices in X. We can add any two chains in an obvious way to obtain another chain by simply following the rule

$$n\sigma + m\sigma = (n+m)\sigma \tag{4.5}$$

Then the collection of all singular n-chains in X becomes a free abelian group, with the empty sum playing the role of the zero-element. The collection of all singular n-simplices is then a basis for this group. We denote this group by $S_n(X)$.

Proposition 1.1.2 ▸ Functoriality

If $f : X \to Y$ is a map of topological spaces, then $\sigma \mapsto f \circ \sigma$ extends to define a graded group homomorphism $f. : S.(X) \to S.(Y)$. Since $(g \circ f) \circ \sigma = g \circ (f \circ \sigma)$, it follows that the association $X \mapsto S.(X)$ defines a functor.

Definition 1.1.3 ▸ Relative Chain Complex

If A is a subspace of X and σ is a singular simplex in A then it can be considered as a singular simplex in X via the inclusion map $A \hookrightarrow X$. In this way, $S_n(A)$ becomes a subgroup of $S_n(X)$. We denote the quotient group $S_n(X)/S_n(A)$ by $S_n(X, A)$.

Remark.

1. $S_n(X, A)$ is a free abelian group with the collection of all singular n-simplexex σ in X whose image is not contained in A as a basis.
2. If $A = \emptyset$, then clearly $S_n(X, A) = S_n(X)$.

Definition 1.1.4

For each integer $r \geq 0$, the ***face map*** or the ***face operator*** $F^r : \mathbb{R}^\infty \to \mathbb{R}^\infty$ is defined by

$$F^r(\mathbf{e}_s) = \begin{cases} \mathbf{e}_s, & \text{if } s < r, \\ \mathbf{e}_{s+1}, & \text{if } s \geq r \end{cases}$$

and extended linearly.

Remark.

1. Note that for each $n \geq r$, F^r carries Δ_{n-1} onto the $(n-1)$-face of Δ_n opposite to the vertex $\mathbf{e}_r$, in an ***order preserving*** manner.
2. We shall denote each F^r restricted to any $\Delta_n, n \geq r$ also by F^r, the exact meaning being understood from the context.

Lemma 1.1.5

$F^r \circ F^{s-1} = F^s \circ F^r$ if $r < s$. [a]

[a] 这里修正了 shastri 书中的一个笔误, 他可能写错了 $s-1$ 的位置

Proof. Since $r < s$, $r \leq s-1$, we fully expand the LHS by definition

$$\begin{aligned} F^r \circ F^{s-1}(\mathbf{e}_k) &= \begin{cases} F^r(\mathbf{e}_k), & k < s-1 \\ F^r(\mathbf{e}_{k+1}), & k \geq s-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < s-1, k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+1}, & s-1 \leq k < r-1 \text{ (empty set)} \\ \mathbf{e}_{k+2}, & k \geq s-1, k \geq r-1 \end{cases} \\ &= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases} \end{aligned}$$

Similarly,

$$F^s \circ F^r(\mathbf{e}_k) = \begin{cases} F^s(\mathbf{e}_k), & k < r \\ F^s(\mathbf{e}_{k+1}), & k \geq r \end{cases}$$
$$= \begin{cases} \mathbf{e}_k, & k < r, k \leq s \\ \mathbf{e}_{k+1}, & k < r, k \geq s \,(\text{empty set}) \\ \mathbf{e}_{k+1}, & k \geq r, k < s-1 \\ \mathbf{e}_{k+2}, & k \geq r, k \geq s-1 \end{cases}$$
$$= \begin{cases} \mathbf{e}_k, & k < r \\ \mathbf{e}_{k+1}, & r \leq k < s-1 \\ \mathbf{e}_{k+2}, & k \geq s-1 \end{cases}$$

□

Definition 1.1.6

For $n \geq 1$ and for any singular n-simplex σ in X, we define

$$\partial_n(\sigma) = \sum_{r=0}^{n} (-1)^r \sigma \circ F^r$$

and extend it linearly to obtain homomorphisms $\partial_n : S_n(X,A) \to S_{n-1}(X,A)$. We define $S_n(X,A) = (0)$ for $n \leq -1$ and $\partial_n = 0$ for $n \geq 0$.

Note.
σ is a map from Δ_n to X. $\sigma \circ F^r$ is the restriction of σ on its face opposite to $\mathbf{e}_r$. Then $\partial_n(\sigma)$ is something like a closed $(n-1)$ - surface consists of σ 's faces.

Proposition 1.1.7 ▶ Singular Chain Complex

$(S(X,A),\partial) = \{S_n(X,A),\partial_n\}$ is a chain complex and the association $(X,A) \rightsquigarrow (S.(X,A),\partial)$ is a functor.

Proof. We shall first show that $\partial^2(\sigma) = 0$ for any singular n-simplex, $n \geq 2$:

$$
\begin{aligned}
\partial^2(\sigma) = \partial\left(\sum_{s=0}^{n} (-1)^s \sigma \circ F^s\right) \\
&= \sum_{s=0}^{n} (-1)^s \partial(\sigma \circ F^s) \\
&= \sum_{s=0}^{n} \sum_{r=0}^{n-1} (-1)^s (-1)^r \sigma \circ F^s \circ F^r \\
&= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} + \sum_{n-1 \geq r \geq s \geq 0} (-1)^{s+r} \sigma \circ F^s \circ F^r \\
&= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \circ F^r \circ F^{s-1} + \sum_{n-1 \geq s \geq r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^s \\
&= \sum_{0 \leq r < s \leq n} (-1)^{s+r} \circ F^r \circ F^{s-1} - \sum_{n \geq s > r \geq 0} (-1)^{s+r} \sigma \circ F^r \circ F^{s-1} \\
&= 0
\end{aligned}
$$

Given a map $f : (X, A) \to (Y, B)$. Since $(f \circ \sigma) \circ F^r = f \circ (\sigma \circ F^r)$, we define $f_.$ as $f_.(\sigma) := f \circ \sigma$ extended by linearlity. Then $\partial(f_.(\sigma)) = f_.(\partial(\sigma))$ by the definition of ∂ and $f_.$'s linearlity. The rest of the details are ommited. □

Definition 1.1.8

The ***relative singular homology groups*** of a topological pair (X, A) are the homology groups $H_*(S_.(X, A))$ and are denoted by $H_*(X, A)$. Taking $A = \varnothing$, we get the homology groups $H_*(X)$.

Remark.
Given any commutative ring R

1. We could have formed the ***free module*** $S_n(X; R)$ of all the singular n-chains in X.
2. Exactly as before, we would have got a chain complex $S_.(X; R)$, sub-chain complex $S_.(A; R)$ and the quotient complex $S_.(X, A; R)$, etc.
3. The corresponding homology groups $H_*(X; R)$, etc. , are then called the homology groups X with coefficients in R. Clearly they form a ***graded*** R***-module*** by themselves. It is customary to drop the notation R when $R = \mathbb{Z}$.

1.2 Basic Properties

Remark.

- It is easily seen that $\{S_\cdot(X,A),\partial\}$ is a functor from the category of topological pairs to the category of chain complexes; thus $H_*(X,A)$ and $H_*(X)$ are also functors.
- Given $f:(X,A)\to(Y,B)$, we shall denote by $f_\cdot$ the map induced by f on $S_\cdot(X,A)$ and by f_* the map induced by f on $H_*(X,A)$.

Lemma 1.2.1

There exist functorial chain homotopies

$$h : S_q(X,A)\to S_{q+1}((X,A)\times\mathbb{I}), q\geq 0,$$

,called ***prism operator*** , between the two inclusion induced maps $\eta_0,\eta_1 : S.(X,A)\to S.((X,A)\times\mathbb{I})$, where for any $t\in\mathbb{I}$, η_t is the inclusion map

$$\eta_t(x)=(x,t), \quad t\in\mathbb{I}$$

Theorem 1.2.2 ▸ Homotopy invariance

Let $f,g:(X,A)\to(Y,B)$ be ***homotopic*** to ***each other***. Then $f_*=g_*$ on $H_*(X,A)$.

Proof. Let $G:(X,A)\times\mathbb{I}\to(Y,B)$ be a homotopy between f and g. For any $t\in\mathbb{I}$, let $\eta_t : (X,A)\to(X,A)\times\mathbb{I}$ be inclusion map $\eta_t(x)=(x,t)$. Then $G\circ\eta_0=f, G\circ\eta_1=g$. Only need to show that chain maps $(\eta_0).$ and $(\eta_1).$ are chain homotopic, thus $(\eta_0)_*=(\eta_1)_*$, $f_*=g_*$, which is implied by the lemma above. □

Corollary 1.2.3

Homotopy-equivalent topological spaces have isomorphic homology groups; in particular, a contractible space has the homology groups of a point-space.

Example 1.2.4

Let $\star$ be a point space. We have

$$H_k(\star) = \begin{cases} \mathbb{Z}, & k = 0 \\ 0, & k \geq 1 \end{cases}$$

Proof. Let n be any positive integer. There is a unique singular n-simplex $\sigma : |\Delta_n| \to \{\star\}$ and hence $S_n(\star) \cong \mathbb{Z}$ for each $n \geq 0$. Observe that $\partial : S_{2n+1}(\star) \to S_{2n}(\star)$ is the zero-map because Δ_{2n+1} has an even number of $(2n)$-faces and the terms in $\partial(\sigma)$ cancel in pairs. However, similar reasoning tells us that $\partial : S_{2n}(\star) \to S_{2n-1}(\star)$ is an isomorphism. Thus we have a chain complex

$$\dots \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\cong} \mathbb{Z} \longrightarrow 0$$

Thus it follows that $H_0(\star) \cong \mathbb{Z}$ and $H_i(\star) = (0)$ for all $i > 0$. □

Example 1.2.5

Let $\{X_j : j \in J\}$ be the set of path connected components of X so that $X = \bigsqcup_{j \in J} X_j$. Then

$$H_*(X) = \bigoplus_{j \in J} H_*(X_j)$$

Remark.
Thus in practice, we can often assume that the space itself is path connected, while dealing with its homology groups.

Proof. Since $|\Delta_n|$ is a path connected space, it follows that any singular n-simplex in X is contained completely in one of the path components X_j. Therefore, it follows that the entire chain complex $S_*(X)$ is the direct sum of the subchain complexes $S_*(X_j)$. Hence the homology is also a direct sum of $H_*(X_j)$, $j \in J$ by Theorem **??**. □

Theorem 1.2.6 ▶ Homology Long Exact Sequence of the pair (X, A)

By definition of $S_.(X, A)$, we have an exact sequence of chain complexes

$$0 \to S_.(A) \to S_.(X) \to S_.(X, A) \to 0.$$

By Theorem ??, this yields the long exact sequence of homology groups

$$\begin{cases} \dots \to H_i(A) \to H_i(X) \to H_i(X, A) \xrightarrow{\delta} H_{i-1}(A) \to \dots \\ \quad \dots \to H_1(X, A) \xrightarrow{\delta} H_0(A) \to H_0(X) \to H_0(X, A), \end{cases} \tag{4.6}$$

which is functorial.

Remark.
This sequence comes handy in many computations as we shall see soon.

1.3 Homology Excision and Mayer-Vietoris

Lemma 1.3.1

Let $X = A \cup B$. Then the following statements are equivalent.

1. $S_.(A) + S_.(B) \to S_.(X)$ induces isomorphisms in homology.
2. $[S_.(A) + S_.(B)]/S_.(B) \to S_.(X)/S_.(B)$ induces isomorphisms in homology.
3. $[S_.(A) + S_.(B)]/S_.(A) \to S_.(X)/S_.(A)$ induces isomorphisms in homology.
4. $S_.(A)/S_.(A \cap B) \to S_.(X)/S_.(B)$ induces isomorphisms in homology.
5. $S_.(B)/S_.(A \cap B) \to S_.(X)/S_.(A)$ induces isomorphisms in homology.

Note.
1,2 等价的好处就在于, 它告诉我们, 我们所需要的同调群的同构对于一部分的信息是不敏感的.

Proof sketch.
对于 1,2 的等价性, 涉及四种同调群, 我们借助 B 的同调将它们相互联系. 用短正合列到长正合列的诱导行为将四种同调群两两联系, 并以 B 的同调为桥梁, 利用诱导行为的函子性将两组同调群联系起来. B 到自身的恒等给出一组同构, 配合已知的一组同构可以利用 Five lemma 夹出另一组同构.

2,4 的等价性直接就是第二同构定理的结果，本质上与同调关系不大.

Proof. The equivalence of (1) and (2) is a consequence of the Five lemma applied to the ladder of homology long exact sequences given by the ladder of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & S.(B) \arrow[r] \arrow[d] & S.(A)+S.(B) \arrow[r] \arrow[d] & {[S.(A)+S.(B)]/S.(B)} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & S.(B) \arrow[r] & S.(X) \arrow[r] & S.(X)/S.(B) \arrow[r] & 0.
\end{tikzcd}$$

The equivalence of (2) and (4) follows from the commutative diagram below in which the horizontal arrow is the isomorphism given by the Noether's isomorphism theorem.

$$\begin{tikzcd}
S.(A)/S.(A\cap B) \arrow[r] \arrow[dr] & {[S.(A)+S.(B)]/S.(B)} \arrow[d] \\
 & S.(X)/S.(B)
\end{tikzcd}$$

□

Definition 1.3.2

Let X be a topological space, A, B be two subspaces such that $X = A\cup B$. We then say that the inclusion map $(A, A\cap B) \hookrightarrow (X, B)$ an excision map. A pair $\{A, B\}$ of subspaces of a space X is called an ***excisive couple*** for singular homology, if the inclusion map

$$S_.(A) + S_.(B) \hookrightarrow S_.(A\cup B)$$

induces isomorphisms in homology or if any of the other four equivalent conditions in the above lemma is satisfied.

Note.
excisive couple 是一对能代表 (X, A) 和 (X, B) 的同调性质的子集 (A, B)，是 X 的一个良好分解. 它使得全空间 X 在切除 $X\setminus A$ 或 $X\setminus B$ 的切除行为上, 在同调的意义下是无损的.

Theorem 1.3.3 ▶ Homology excision

The inclusion map $(A, A \cap B) \hookrightarrow (A \cup B, B)$ of an excisive couple $\{A, B\}$ induces an isomorphism of homology groups $H_*(A, A \cap B) \to H_*(A \cup B, B)$.

Theorem 1.3.4

If $X = X_1 \cup X_2 = \mathrm{int}_X(X_1) \cup \mathrm{int}_X(X_2)$ then, $\{X_1, X_2\}$ is an excisive couple for the singular homology.

Note.
也就是说，当分解的一个组件的边界能被另一个组件的内部“吃掉”时，这组分解就是切除对.

Example 1.3.5

Here are a few examples of excisive couples which occur in practice.

1. $\{\mathbb{D}^n, \mathbb{R}^n \setminus \{0\}\}$ is an excisive couple and hence the inclusion map $(\mathbb{D}^n, \mathbb{D}^n \setminus \{0\}) \hookrightarrow (\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ will induce isomorphisms in all singular homology groups.
2. Let N denote the north pole on $\mathbb{S}^n$ and $\bar{U}$ denote the closed upper hemisphere. Then $\{\mathbb{S}^n \setminus \{N\}, \bar{U}\}$ is an excisive couple, which gives isomorphisms of $H_*(\bar{U}, \bar{U} \setminus \{N\}) \to H_*(\mathbb{S}^n, \mathbb{S}^n \setminus \{N\})$.

Theorem 1.3.6

Given subsets X_1 and X_2 of a topological space X, let us denote the inclusion maps of X_i in to X by $\eta_i, i = 1, 2$. They induce inclusion maps of the chain complexes which we shall denote again by $\eta_i : S.(X_i) \to S.(X), i = 1, 2$. Now consider the chain map

$$(\eta_1, -\eta_2) : S.(X_1) \oplus S.(X_2) \to S.(X).$$

The submodule generated by $S.(X_1)$ and $S.(X_2)$ in $S.(X)$ is denoted by $S.(X_1)+S.(X_2)$. Then we have a short exact sequence of chain complexes

$$0 \to S.(X_1 \cap X_2) \to S.(X_1) \oplus S.(X_2) \to S.(X_1) + S.(X_2) \to 0$$

If $\{X_1, X_2\}$ is an excisive couple, then we can obtain the ***Mayer-Vietoris***

sequence:

$$\left.\begin{array}{l}\cdots \to H_{i+1}(X_1 \cup X_2) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(X_1 \cup X_2) \to \cdots \\ \cdots \to H_1(X_1 \cup X_2) \xrightarrow{\partial} H_0(X_1 \cap X_2) \to H_0(X_1) \oplus H_0(X_2) \to H_0(X_1 \cup X_2)\end{array}\right\}$$

with

$$i_*(z) = (i_{1*}z, -i_{2*}z); \quad j_*(z_1, z_2) = j_{1*}z_1 + j_{2*}z_2$$

where,

$$\begin{array}{ll} i_1 : X_1 \cap X_2 \hookrightarrow X_1, & i_2 : X_1 \cap X_2 \hookrightarrow X_2 \\ j_1 : X_1 \hookrightarrow X_1 \cup X_2, & j_2 : X_2 \hookrightarrow X_1 \cup X_2 \end{array}$$

are inclusion maps.

Remark.
The Mayer-Vietoris sequence is one of the major ready-to-use-tools in computing homology.

Proof. The kernal of $(\eta_1, -\eta_2)$ is $S.(X_1) \cap S.(X_2) = S.(X_1 \cap X_2)$. The image is clearly $S.(X_1) + S.(X_2)$. Thus the short sequence is exact. Then we obtain the long exact sequence

$$\begin{array}{l}\cdots \to H_{i+1}(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_i(X_1 \cap X_2) \xrightarrow{i_*} H_i(X_1) \oplus H_i(X_2) \xrightarrow{j_*} H_i(S.(X_1) + S.(X_2)) \\ \cdots \to H_1(S.(X_1) + S.(X_2)) \xrightarrow{\partial} H_0(X_1 \cap X_2) \to H_0(X_1) \oplus H_0(X_2) \to H_0(S.(X_1) + S.(X_2))\end{array}$$

By the definition of excisive couple , we can replace $H_i(S.(X_1) + S.(X_2))$ by $H_i(X_1 \cup X_2)$ to obtain Mayer-Vietoris. □